KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5052 | 5 Credits: 4 | Program Core | Source-grounded

Vehicle Diagnostics and Service.

 To provide knowledge on diagnostic and maintenance procedure for automobile

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5052 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5052.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Automobile Engineering
Course code	5052
Course title	Vehicle Diagnostics and Service.
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide knowledge on diagnostic and maintenance procedure for automobile
- To get familiarized to logical diagnostic processes and practices and acquire guidance

Course outcomes

CO	Official outcome	Study level
CO1	Make use of the methods of engine reconditioning. 15 Applying Identify the diagnostics of fuel and electrical	See official syllabus
CO2	15 Understanding systems Summarize the diagnostics of transmission and drive	See official syllabus
CO3	14 Understanding line components. Identify the diagnostic procedures for suspension,	See official syllabus
CO4	14 Understanding steering and brake systems.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Make use of the methods of engine reconditioning	5	As specified
Module 2	Identify the diagnostics of fuel and electrical systems	4	As specified
Module 3	Summarize the diagnostics of transmission and drive line components.	4	As specified
Module 4	systems.	4	As specified

MODULE 1

Make use of the methods of engine reconditioning

This module is presented from the official syllabus scope for Vehicle Diagnostics and Service.. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Make use of the methods of engine reconditioning
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Explain engine decarbonizing methods

Explain engine decarbonizing methods is an official topic in Module 1 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain engine decarbonizing methods using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain engine decarbonizing methods should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain engine decarbonizing methods means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain engine decarbonizing methods □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Summarize the basic engine tests.

Summarize the basic engine tests. is an official topic in Module 1 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the basic engine tests. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the basic engine tests. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the basic engine tests. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Summarize the basic engine tests. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Identify the methods for engine dismantling.

Identify the methods for engine dismantling. is an official topic in Module 1 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the methods for engine dismantling. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the methods for engine dismantling. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the methods for engine dismantling. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Identify the methods for engine dismantling. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Explain servicing of various engine components.

Explain servicing of various engine components. is an official topic in Module 1 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain servicing of various engine components. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain servicing of various engine components. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain servicing of various engine components. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain servicing of various engine components. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Identify the steps in engine assembling 2 Applying Engine decarbonizing methods-mechanical and chemical, engine condition diagnosis - engine tests - dry compression test, wet compression test, oil pressure test, vacuum test, cylinder leakage test, exhaust restriction test, engine removal and dis - assembly - removal of engine from vehicle, dismantling of engine. Cylinder head reconditioning - head inspection, head resurfacing, valve seat refacing, valve guide reconditioning, valve spring inspection, valve reconditioning, cylinder block reconditioning - align boring of saddle bore, block surface reconditioning, cylinder boring and honing, crankshaft servicing, camshaft servicing, piston and connecting rod servicing, flywheel reconditioning, Engine assembly - preparation for assembly, steps for assembling, testing on dynamometer, engine installation, break - in.

Identify the steps in engine assembling 2 Applying Engine decarbonizing methods-mechanical and chemical, engine condition diagnosis - engine tests - dry compression test, wet compression test, oil pressure test, vacuum test, cylinder leakage test, exhaust restriction test, engine removal and dis - assembly - removal of engine from vehicle, dismantling of engine. Cylinder head reconditioning - head inspection, head resurfacing, valve seat refacing, valve guide reconditioning, valve spring inspection, valve reconditioning, cylinder block reconditioning - align boring of saddle bore, block surface reconditioning, cylinder boring and honing, crankshaft servicing, camshaft servicing, piston and connecting rod servicing, flywheel reconditioning, Engine assembly - preparation for assembly, steps for assembling, testing on dynamometer, engine installation, break - in. is an official topic in Module 1 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify the steps in engine assembling 2 Applying Engine decarbonizing methods-mechanical and chemical, engine condition diagnosis - engine tests - dry compression test, wet compression test, oil pressure test, vacuum test, cylinder leakage test, exhaust restriction test, engine removal and dis - assembly - removal of engine from vehicle, dismantling of engine. Cylinder head reconditioning - head inspection, head resurfacing, valve seat refacing, valve guide reconditioning, valve spring inspection, valve reconditioning, cylinder block reconditioning - align boring of saddle bore, block surface reconditioning, cylinder boring and honing, crankshaft servicing, camshaft servicing, piston and connecting rod servicing, flywheel reconditioning, Engine assembly - preparation for assembly, steps for assembling, testing on dynamometer, engine installation, break - in. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify the steps in engine assembling 2 applying engine decarbonizing methods-mechanical and chemical, engine condition diagnosis - engine tests - dry compression test, wet compression test, oil pressure test, vacuum test, cylinder leakage test, exhaust restriction test, engine removal and dis - assembly - removal of engine from vehicle, dismantling of engine. cylinder head reconditioning - head inspection, head resurfacing, valve seat refacing, valve guide reconditioning, valve spring inspection, valve reconditioning, cylinder block reconditioning - align boring of saddle bore, block surface reconditioning, cylinder boring and honing, crankshaft servicing, camshaft servicing, piston and connecting rod servicing, flywheel reconditioning, engine assembly - preparation for assembly, steps for assembling, testing on dynamometer, engine installation, break - in. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify the steps in engine assembling 2 Applying Engine decarbonizing methods-mechanical and chemical, engine condition diagnosis - engine tests - dry compression test, wet compression test, oil pressure test, vacuum test, cylinder leakage test, exhaust restriction test, engine removal and dis - assembly - removal of engine from vehicle, dismantling of engine. Cylinder head reconditioning - head inspection, head resurfacing, valve seat refacing, valve guide reconditioning, valve spring inspection, valve reconditioning, cylinder block reconditioning - align boring of saddle bore, block surface reconditioning, cylinder boring and honing, crankshaft servicing, camshaft servicing, piston and connecting rod servicing, flywheel reconditioning, Engine assembly - preparation for assembly, steps for assembling, testing on dynamometer, engine installation, break - in. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Identify the steps in engine assembling 2 Applying Engine decarbonizing methods-mechanical and chemical, engine condition diagnosis - engine tests - dry compression test, wet compression test, oil pressure test, vacuum test, cylinder leakage test, exhaust restriction test, engine removal and dis - assembly - removal of engine from vehicle, dismantling of engine. Cylinder head reconditioning - head inspection, head resurfacing, valve seat refacing, valve guide reconditioning, valve spring inspection, valve reconditioning, cylinder block reconditioning - align boring of saddle bore, block surface reconditioning, cylinder boring and honing, crankshaft servicing, camshaft servicing, piston and connecting rod servicing, flywheel reconditioning, Engine assembly - preparation for assembly, steps for assembling, testing on dynamometer, engine installation, break - in. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Identify the diagnostics of fuel and electrical systems

This module is presented from the official syllabus scope for Vehicle Diagnostics and Service.. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Identify the diagnostics of fuel and electrical systems
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding system Utilize the servicing procedure of ignition

2 Understanding system Utilize the servicing procedure of ignition is an official topic in Module 2 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding system Utilize the servicing procedure of ignition using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding system utilize the servicing procedure of ignition should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Understanding system Utilize the servicing procedure of ignition means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding system Utilize the servicing procedure of ignition □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 5 Applying system components Demonstrate the servicing of fuel system

5 Applying system components Demonstrate the servicing of fuel system is an official topic in Module 2 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Applying system components Demonstrate the servicing of fuel system using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 applying system components demonstrate the servicing of fuel system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 5 Applying system components Demonstrate the servicing of fuel system means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 5 Applying system components Demonstrate the servicing of fuel system [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 6 Understanding components Identify the inspection of emission control

6 Understanding components Identify the inspection of emission control is an official topic in Module 2 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding components Identify the inspection of emission control using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding components identify the inspection of emission control should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 6 Understanding components Identify the inspection of emission control means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 6 Understanding components Identify the inspection of emission control □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ 2 Applying systems On-board diagnostic systems-monitors and self-diagnosis, MIL and DLC, scan tools, basic diagnosis of electronic engine control systems. Engine misfiring causes, visual inspection of ignition system, primary circuit checking, secondary circuit checking, ignition coil checking, pickup coil checking, distributor cap and rotor checking, coil over plug checking, spark plug inspection, no start diagnosis of EI system, usage of engine analyzer, ignition timing setting. Preliminary checks for fuel system, fuel system possible faults and remedy, fuel system testing-fuel pressure testing, fuel delivery testing, pressure regulator diagnosis, air induction system checking, sensor diagnosis, throttle body servicing, injector servicing, IAC inspection and service, Diesel injection fault diagnosis, basic inspection of emission control systems.

2 Applying systems On-board diagnostic systems-monitors and self-diagnosis, MIL and DLC, scan tools, basic diagnosis of electronic engine control systems. Engine misfiring causes, visual inspection of ignition system, primary circuit checking, secondary circuit checking, ignition coil checking, pickup coil checking, distributor cap and rotor checking, coil over plug checking, spark plug inspection, no start diagnosis of EI system, usage of engine analyzer, ignition timing setting. Preliminary checks for fuel system, fuel system possible faults and remedy, fuel system testing-fuel pressure testing, fuel delivery testing, pressure regulator diagnosis, air induction system checking, sensor diagnosis, throttle body servicing, injector servicing, IAC inspection and service, Diesel injection fault diagnosis, basic inspection of emission control systems. is an official topic in Module 2 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying systems On-board diagnostic systems-monitors and self-diagnosis, MIL and DLC, scan tools, basic diagnosis of electronic engine control systems. Engine misfiring causes, visual inspection of ignition system, primary circuit checking, secondary circuit checking, ignition coil checking, pickup coil checking, distributor cap and rotor checking, coil over plug checking, spark plug inspection, no start diagnosis of EI system, usage of engine analyzer, ignition timing setting. Preliminary checks for fuel system, fuel system possible faults and remedy, fuel system testing-fuel pressure testing, fuel delivery testing, pressure regulator diagnosis, air induction system checking, sensor diagnosis, throttle body servicing, injector servicing, IAC inspection and service, Diesel injection fault diagnosis, basic inspection of emission control systems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying systems on-board diagnostic systems-monitors and self-diagnosis, mil and dlc, scan tools, basic diagnosis of electronic engine control systems. engine misfiring causes, visual inspection of ignition system, primary circuit checking, secondary circuit checking, ignition coil checking, pickup coil checking, distributor cap and rotor checking, coil over plug checking, spark plug inspection, no start diagnosis of ei system, usage of engine analyzer, ignition timing setting. preliminary checks for fuel system, fuel system possible faults and remedy, fuel system testing-fuel pressure testing, fuel delivery testing, pressure regulator diagnosis, air induction system checking, sensor diagnosis, throttle body servicing, injector servicing, iac inspection and service, diesel injection fault diagnosis, basic inspection of emission control systems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying systems On-board diagnostic systems-monitors and self-diagnosis, MIL and DLC, scan tools, basic diagnosis of electronic engine control systems. Engine misfiring causes, visual inspection of ignition system, primary circuit checking, secondary circuit checking, ignition coil checking, pickup coil checking, distributor cap and rotor checking, coil over plug checking, spark plug inspection, no start diagnosis of EI system, usage of engine analyzer, ignition timing setting. Preliminary checks for fuel system, fuel system possible faults and remedy, fuel system testing-fuel pressure testing, fuel delivery testing, pressure regulator diagnosis, air induction system checking, sensor diagnosis, throttle body servicing, injector servicing, IAC inspection and service, Diesel injection fault diagnosis, basic inspection of emission control systems. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Applying systems On-board diagnostic systems-monitors and self-diagnosis, MIL and DLC, scan tools, basic diagnosis of electronic engine control systems. Engine misfiring causes, visual inspection of ignition system, primary circuit checking, secondary circuit checking, ignition coil checking, pickup coil checking, distributor cap and rotor checking, coil over plug checking, spark plug inspection, no start diagnosis of EI system, usage of engine analyzer, ignition timing setting. Preliminary checks for fuel system, fuel system possible faults and remedy, fuel system testing-fuel pressure testing, fuel delivery testing, pressure regulator diagnosis, air induction system checking, sensor diagnosis, throttle body servicing, injector servicing, IAC inspection and service, Diesel injection fault diagnosis, basic inspection of emission control systems. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Summarize the diagnostics of transmission and drive line components.

This module is presented from the official syllabus scope for Vehicle Diagnostics and Service.. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Summarize the diagnostics of transmission and drive line components.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▾ Explain the service procedures of clutch. 3 Understanding Outline the troubles and remedies of

Explain the service procedures of clutch. 3 Understanding Outline the troubles and remedies of is an official topic in Module 3 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the service procedures of clutch. 3 Understanding Outline the troubles and remedies of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the service procedures of clutch. 3 understanding outline the troubles and remedies of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Explain the service procedures of clutch. 3 Understanding Outline the troubles and remedies of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Explain the service procedures of clutch. 3 Understanding Outline the troubles and remedies of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▼ 4 Understanding transmission system. Demonstrate the servicing of propeller shaft and

4 Understanding transmission system. Demonstrate the servicing of propeller shaft and is an official topic in Module 3 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding transmission system. Demonstrate the servicing of propeller shaft and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding transmission system. demonstrate the servicing of propeller shaft and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding transmission system. Demonstrate the servicing of propeller shaft and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding transmission system. Demonstrate the servicing of propeller shaft and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 4 Understanding differential. Summarize the methods for tyre rotation and

4 Understanding differential. Summarize the methods for tyre rotation and is an official topic in Module 3 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding differential. Summarize the methods for tyre rotation and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding differential. summarize the methods for tyre rotation and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 4 Understanding differential. Summarize the methods for tyre rotation and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding differential. Summarize the methods for tyre rotation and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

▼ 3 Understanding wheel balancing. Clutch maintenance- clutch linkage adjustment, clutch linkage lubrication, clutch problem diagnosis - slippage, dragging and binding, chatter, pedal pulsation, vibration, clutch service. Transmission and transaxle removal and refitting, transmission possible faults and remedy, dismantling of propeller shaft and universal joint, defects in propeller shaft, servicing of differential and rear axle, differential troubles and adjustments, causes of tyre wear, tyre rotation, wheel balancing. Identify the diagnostic procedures for suspension, steering and brake

3 Understanding wheel balancing. Clutch maintenance- clutch linkage adjustment, clutch linkage lubrication, clutch problem diagnosis - slippage, dragging and binding, chatter, pedal pulsation, vibration, clutch service. Transmission and transaxle removal and refitting, transmission possible faults and remedy, dismantling of propeller shaft and universal joint, defects in propeller shaft, servicing of differential and rear axle, differential troubles and adjustments, causes of tyre wear, tyre rotation, wheel balancing. Identify the diagnostic procedures for suspension, steering and brake is an official topic in Module 3 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding wheel balancing. Clutch maintenance- clutch linkage adjustment, clutch linkage lubrication, clutch problem diagnosis - slippage, dragging and binding, chatter, pedal pulsation, vibration, clutch service. Transmission and transaxle removal and refitting, transmission possible faults and remedy, dismantling of propeller shaft and universal joint, defects in propeller shaft, servicing of differential and rear axle, differential troubles and adjustments, causes of tyre wear, tyre rotation, wheel balancing. Identify the diagnostic procedures for suspension, steering and brake using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding wheel balancing. clutch maintenance- clutch linkage adjustment, clutch linkage lubrication, clutch problem diagnosis - slippage, dragging and binding, chatter, pedal pulsation, vibration, clutch service. transmission and transaxle removal and refitting, transmission possible faults and remedy, dismantling of propeller shaft and universal joint, defects in propeller shaft, servicing of differential and rear axle, differential troubles and adjustments, causes of tyre wear, tyre rotation, wheel balancing. identify the diagnostic procedures for suspension, steering and brake should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding wheel balancing. Clutch maintenance- clutch linkage adjustment, clutch linkage lubrication, clutch problem diagnosis - slippage, dragging and binding, chatter, pedal pulsation, vibration, clutch service. Transmission and transaxle removal and refitting, transmission possible faults and remedy, dismantling of propeller shaft and universal joint, defects in propeller shaft, servicing of differential and rear axle, differential troubles and adjustments, causes of tyre wear, tyre rotation, wheel balancing. Identify the diagnostic procedures for suspension, steering and brake means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding wheel balancing. Clutch maintenance- clutch linkage adjustment, clutch linkage lubrication, clutch problem diagnosis - slippage, dragging and binding, chatter, pedal pulsation, vibration, clutch service. Transmission and transaxle removal and refitting, transmission possible faults and remedy, dismantling of propeller shaft and universal joint, defects in propeller shaft, servicing of differential and rear axle, differential troubles and adjustments, causes of tyre wear, tyre rotation, wheel balancing. Identify the diagnostic procedures for suspension, steering and brake □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

systems.

This module is presented from the official syllabus scope for Vehicle Diagnostics and Service.. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: systems.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Applying diagnosis. Make use of the procedure for MacPherson strut

3 Applying diagnosis. Make use of the procedure for MacPherson strut is an official topic in Module 4 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying diagnosis. Make use of the procedure for MacPherson strut using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying diagnosis. make use of the procedure for macpherson strut should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Applying diagnosis. Make use of the procedure for MacPherson strut means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Applying diagnosis. Make use of the procedure for MacPherson strut □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 1 Applying replacement.

1 Applying replacement. is an official topic in Module 4 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Applying replacement. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 applying replacement. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Applying replacement. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 1 Applying replacement. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Outline the servicing of steering system.

Outline the servicing of steering system. is an official topic in Module 4 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the servicing of steering system. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the servicing of steering system. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Outline the servicing of steering system. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Outline the servicing of steering system. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Explain the servicing procedure of brake system. 5 Understanding Suspension system diagnosis, front suspension component servicing - coil springs, torsion bar, ball joints, bushings, shock absorbers, removal and refitting of MacPherson strut, rear suspension component servicing - solid rear axle, semi-independent and independent, basic diagnosis of electronic suspension. Steering system diagnosis - common complaints and reasons, tyre wear pattern and steering troubles, steering gear adjustments, rack and pinion service, steering system component servicing, power steering system servicing, checking of wheel alignment. Drum brake diagnosis - drum brake disassembly, brake lining inspection, wheel cylinder inspection and service, drum brake adjustment, disk brake diagnosis-visual inspection, caliper service, master cylinder servicing, brake bleeding procedure, basic diagnosis procedure for ABS. Text / Reference: T/R Book Title/Author Automotive Technology: A Systems Approach by Jack Erjavec, Cengage T1 Learning Automotive Technology: Principles, Diagnosis, and Service by James D. R1 Halderman, Prentice Hall Advanced Automotive Fault Diagnosis, Automotive Technology: Vehicle R2 Maintenance and Repair, Tom Denton, Routledge R3 Automobile Engg vol 1 and II, Kirpal Singh, Standard Publication R4 Automotive Mechanics, Anglin And Crouse, McGraw-Hill Online Resources: Sl.No Website Link 1 https://www.wikihow.com/Rebuild-an-Engine 2 https://drivinglife.net/how-to-use-automotive-scan-tools/ 3 https://www.youtube.com/watch?v=gPotEQDJhsk

Explain the servicing procedure of brake system. 5 Understanding Suspension system diagnosis, front suspension component servicing - coil springs, torsion bar, ball joints, bushings, shock absorbers, removal and refitting of MacPherson strut, rear suspension component servicing - solid rear axle, semi-independent and independent, basic diagnosis of electronic suspension. Steering system diagnosis - common complaints and reasons, tyre wear pattern and steering troubles, steering gear adjustments, rack and pinion service, steering system component servicing, power steering system servicing, checking of wheel alignment. Drum brake diagnosis -

drum brake disassembly, brake lining inspection, wheel cylinder inspection and service, drum brake adjustment, disk brake diagnosis-visual inspection, caliper service, master cylinder servicing, brake bleeding procedure, basic diagnosis procedure for ABS. Text / Reference: T/R Book Title/Author Automotive Technology: A Systems Approach by Jack Erjavec, Cengage T1 Learning Automotive Technology: Principles, Diagnosis, and Service by James D. R1 Halderman, Prentice Hall Advanced Automotive Fault Diagnosis, Automotive Technology: Vehicle R2 Maintenance and Repair, Tom Denton, Routledge R3 Automobile Engg vol 1 and II, Kirpal Singh, Standard Publication R4 Automotive Mechanics, Anglin And Crouse, McGraw-Hill Online Resources: Sl.No Website Link 1 https://www.wikihow.com/Rebuild-an-Engine 2 https://drivinglife.net/how-to-use-automotive-scan-tools/ 3 https://www.youtube.com/watch?v=gPotEQDJhsk is an official topic in Module 4 of Vehicle Diagnostics and Service.. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the servicing procedure of brake system. 5 Understanding Suspension system diagnosis, front suspension component servicing - coil springs, torsion bar, ball joints, bushings, shock absorbers, removal and refitting of MacPherson strut, rear suspension component servicing - solid rear axle, semi-independent and independent, basic diagnosis of electronic suspension. Steering system diagnosis - common complaints and reasons, tyre wear pattern and steering troubles, steering gear adjustments, rack and pinion service, steering system component servicing, power steering system servicing, checking of wheel alignment. Drum brake diagnosis - drum brake disassembly, brake lining inspection, wheel cylinder inspection and service, drum brake adjustment, disk brake diagnosis-visual inspection, caliper service, master cylinder servicing, brake bleeding procedure, basic diagnosis procedure for ABS. Text / Reference: T/R Book Title/Author Automotive Technology: A Systems Approach by Jack Erjavec, Cengage T1 Learning Automotive Technology: Principles, Diagnosis, and Service by James D. R1 Halderman, Prentice Hall Advanced Automotive Fault Diagnosis, Automotive Technology: Vehicle R2 Maintenance and Repair, Tom Denton, Routledge R3 Automobile Engg vol 1 and II, Kirpal Singh, Standard Publication R4 Automotive Mechanics, Anglin And Crouse, McGraw-Hill Online Resources: Sl.No Website Link 1 https://www.wikihow.com/Rebuild-an-Engine 2 https://drivinglife.net/how-to-use-automotive-scan-tools/ 3 https://www.youtube.com/watch?v=gPotEQDJhsk using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the servicing procedure of brake system. 5 understanding suspension system diagnosis, front suspension component servicing - coil springs, torsion bar, ball joints, bushings, shock absorbers, removal and refitting of macpherson strut, rear suspension component servicing - solid rear axle, semi-independent and independent, basic diagnosis of electronic suspension. steering system diagnosis - common complaints and reasons, tyre wear pattern and steering troubles, steering gear adjustments, rack and pinion service, steering system component servicing, power steering system servicing, checking of wheel alignment. drum brake diagnosis - drum brake disassembly, brake lining inspection, wheel cylinder inspection and service, drum brake adjustment, disk brake diagnosis-visual inspection, caliper service, master cylinder servicing, brake bleeding procedure, basic diagnosis procedure for abs. text / reference: t/r book title/author automotive technology: a systems approach by jack erjavec, cengage t1 learning automotive technology: principles, diagnosis, and service by james d. r1 halderman, prentice hall advanced automotive fault diagnosis, automotive technology: vehicle r2 maintenance and repair, tom denton, routledge r3 automobile engg vol 1 and ii, kirpal singh, standard publication r4 automotive mechanics, anglin and crouse, mcgraw-hill online resources: sl.no website link 1 https://www.wikihow.com/rebuild-an-engine 2 https://drivinglife.net/how-to-use-automotive-scan-tools/ 3 https://www.youtube.com/watch?v=gpoteqdjhsk should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the servicing procedure of brake system. 5 Understanding Suspension system diagnosis, front suspension component servicing - coil springs, torsion bar, ball joints, bushings, shock absorbers, removal and refitting of MacPherson strut, rear suspension component servicing - solid rear axle, semi-independent and independent, basic diagnosis of electronic suspension. Steering system diagnosis - common complaints and reasons, tyre wear pattern and steering troubles, steering gear adjustments, rack and pinion service, steering system component servicing, power steering system servicing, checking of wheel alignment. Drum brake diagnosis - drum brake disassembly, brake lining inspection, wheel cylinder inspection and service, drum brake adjustment, disk brake diagnosis-visual inspection, caliper service, master cylinder servicing, brake bleeding procedure, basic diagnosis procedure for ABS. Text / Reference: T/R Book Title/Author Automotive Technology: A Systems Approach by Jack Erjavec, Cengage T1 Learning Automotive Technology: Principles, Diagnosis, and Service by James D. R1 Halderman, Prentice Hall Advanced Automotive Fault Diagnosis, Automotive Technology: Vehicle R2 Maintenance and Repair, Tom Denton, Routledge R3 Automobile Engg vol 1 and II, Kirpal Singh, Standard Publication R4 Automotive Mechanics, Anglin And Crouse, McGraw-Hill Online Resources: Sl.No Website Link 1 https://www.wikihow.com/Rebuild-an-Engine 2 https://drivinglife.net/how-to-use-automotive-scan-tools/ 3 https://www.youtube.com/watch?v=gPotEQDJhsk means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain the servicing procedure of brake system. 5 Understanding Suspension system diagnosis, front suspension component servicing - coil springs, torsion bar, ball joints, bushings, shock absorbers, removal and refitting of MacPherson strut, rear suspension component servicing - solid rear axle, semi-independent and independent, basic diagnosis of electronic suspension. Steering system diagnosis - common complaints and reasons, tyre wear pattern and steering troubles, steering gear adjustments, rack and pinion service, steering system component servicing, power steering system servicing, checking of wheel alignment. Drum brake diagnosis - drum brake disassembly, brake lining inspection, wheel cylinder inspection and service, drum brake adjustment, disk brake diagnosis-visual inspection, caliper service, master cylinder servicing, brake bleeding procedure, basic diagnosis procedure for ABS. Text / Reference: T/R Book Title/Author Automotive Technology: A Systems Approach by Jack Erjavec, Cengage T1 Learning Automotive Technology: Principles, Diagnosis, and Service by James D. R1 Halderman, Prentice Hall Advanced Automotive Fault Diagnosis, Automotive Technology: Vehicle R2 Maintenance and Repair, Tom Denton, Routledge R3 Automobile Engg vol 1 and II, Kirpal Singh, Standard Publication R4 Automotive Mechanics, Anglin And Crouse, McGraw-Hill Online Resources: Sl.No Website Link 1 https://www.wikihow.com/Rebuild-an-Engine 2 https://drivinglife.net/how-to-use-automotive-scan-tools/ 3 https://www.youtube.com/watch?v=gPotEQDJhsk □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain Explain engine decarbonizing methods. (3 marks)**

▶ **Define or explain Summarize the basic engine tests.. (3 marks)**

▶ **Define or explain Identify the methods for engine dismantling.. (3 marks)**

▶ **Define or explain Explain servicing of various engine components.. (3 marks)**

Module 2

▶ **Define or explain 2 Understanding system Utilize the servicing procedure of ignition. (3 marks)**

▶ **Define or explain 5 Applying system components Demonstrate the servicing of fuel system. (3 marks)**

▶ **Define or explain 6 Understanding components Identify the inspection of emission control. (3 marks)**

▶ **Define or explain 2 Applying systems On-board diagnostic systems- monitors and self-diagnosis, MIL and DLC, scan tools, basic diagnosis of electronic engine control systems. Engine misfiring causes, visual**

inspection of ignition system, primary circuit checking, secondary circuit checking, ignition coil checking, pickup coil checking, distributor cap and rotor checking, coil over plug checking, spark plug inspection, no start diagnosis of EI system, usage of engine analyzer, ignition timing setting. Preliminary checks for fuel system, fuel system possible faults and remedy, fuel system testing-fuel pressure testing, fuel delivery testing, pressure regulator diagnosis, air induction system checking, sensor diagnosis, throttle body servicing, injector servicing, IAC inspection and service, Diesel injection fault diagnosis, basic inspection of emission control systems.. (3 marks)

Module 3

▶ **Define or explain Explain the service procedures of clutch. 3 Understanding Outline the troubles and remedies of. (3 marks)**

▶ **Define or explain 4 Understanding transmission system. Demonstrate the servicing of propeller shaft and. (3 marks)**

▶ **Define or explain 4 Understanding differential. Summarize the methods for tyre rotation and. (3 marks)**

▶ **Define or explain 3 Understanding wheel balancing. Clutch maintenance- clutch linkage adjustment, clutch linkage lubrication, clutch problem diagnosis - slippage, dragging and binding, chatter, pedal pulsation, vibration, clutch service. Transmission and transaxle removal and refitting, transmission possible faults and remedy, dismantling of propeller shaft and universal joint, defects in propeller shaft, servicing of differential and rear axle, differential troubles and adjustments, causes of tyre wear, tyre rotation, wheel balancing. Identify the diagnostic procedures for suspension, steering and brake. (3 marks)**

Module 4

▶ **Define or explain 3 Applying diagnosis. Make use of the procedure for MacPherson strut. (3 marks)**

▶ **Define or explain 1 Applying replacement.. (3 marks)**

▶ **Define or explain Outline the servicing of steering system.. (3 marks)**

▶ **Define or explain Explain the servicing procedure of brake system. 5 Understanding Suspension system diagnosis, front suspension component servicing - coil springs, torsion bar, ball joints, bushings, shock absorbers, removal and refitting of MacPherson strut, rear suspension component servicing - solid rear axle, semi-independent and independent, basic diagnosis of electronic suspension. Steering system diagnosis - common complaints and reasons, tyre wear pattern and steering troubles, steering gear adjustments, rack and pinion service, steering system component servicing, power steering system servicing, checking of wheel alignment. Drum brake diagnosis - drum brake disassembly, brake lining inspection, wheel cylinder inspection and service, drum brake adjustment, disk brake diagnosis-visual inspection, caliper service, master cylinder servicing, brake bleeding procedure, basic diagnosis procedure for ABS. Text / Reference: T/R Book Title/Author Automotive Technology: A Systems Approach by Jack Erjavec, Cengage T1 Learning Automotive Technology: Principles, Diagnosis, and Service by James D. R1 Halderman, Prentice Hall Advanced Automotive Fault Diagnosis, Automotive Technology: Vehicle R2 Maintenance and Repair, Tom Denton, Routledge R3 Automobile Engg vol 1 and II, Kirpal Singh, Standard Publication R4 Automotive Mechanics, Anglin And Crouse, McGraw-Hill Online Resources: Sl.No Website Link 1 https://www.wikihow.com/Rebuild-an-Engine 2 https://drivinglife.net/how-to-use-automotive-scan-tools/ 3 https://www.youtube.com/watch?v=gPotEQDJhsk. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain engine decarbonizing methods**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding system Utilize the servicing procedure of ignition**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain the service procedures of clutch. 3 Understanding Outline the troubles and remedies of**.

Module 4

1-mark definition: State the meaning of **3 Applying diagnosis. Make use of the procedure for MacPherson strut**.

3-mark explanation: Explain one principle, process or comparison from

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://www.wikihow.com/Rebuild-an-Engine
- https://drivinglife.net/how-to-use-automotive-scan-tools/ https://www.youtube.com/watch?v=gPotEQDJhsk

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 5052. Verify institutional updates against the current official curriculum.